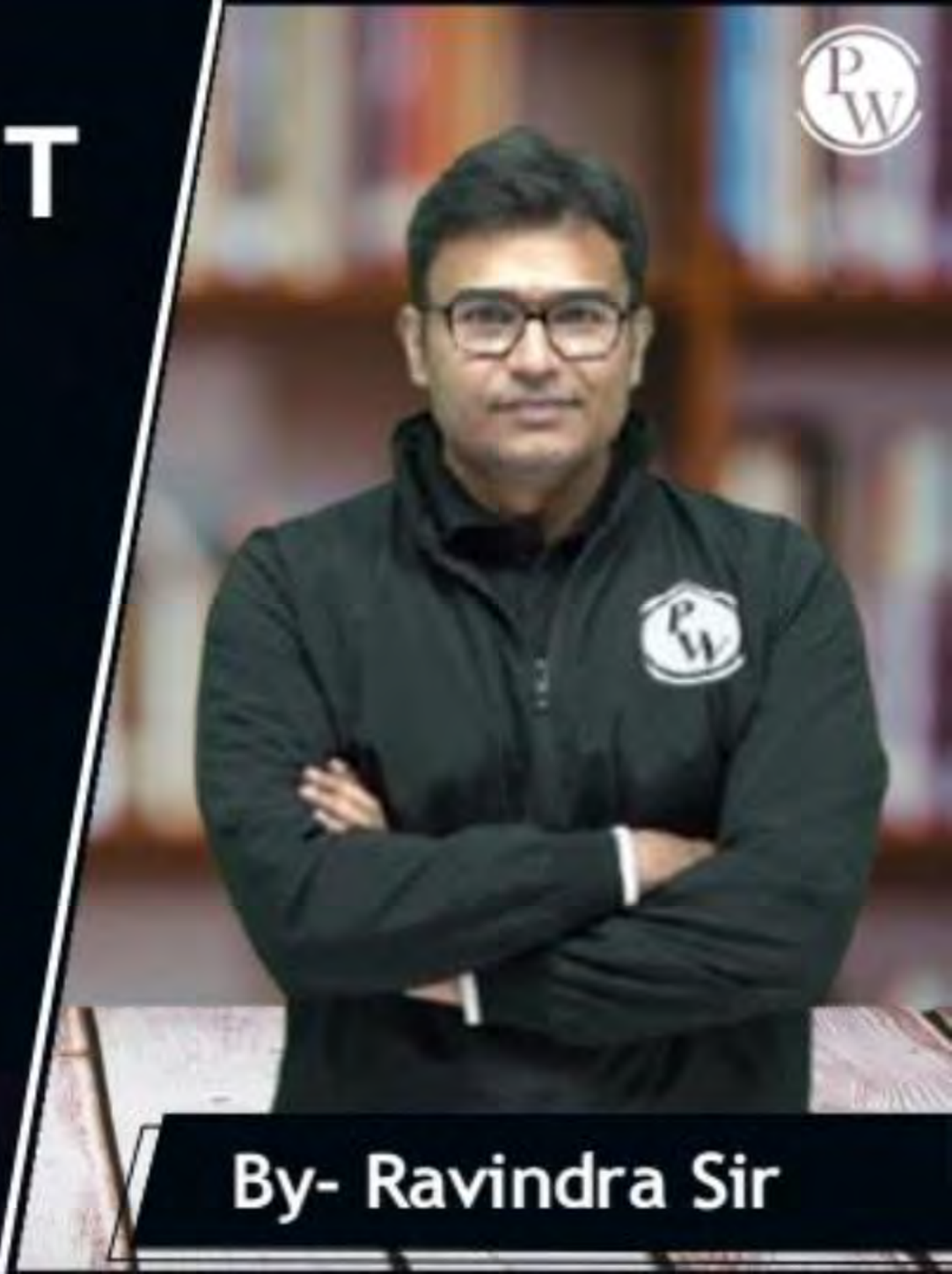
Computer Science & IT

COMPILER DESIGN
(CD)

Code Optimization

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

CO

Topics to be Covered



- Topic
- Topic
- Topic

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

Loop optimizations :-

1) loop invariant code motion (LICM) & frequency reduction

→ moves computations that produce same result on every iteration outside the loop

```
for(int i=0; i<n; i++)
{ x=a*b;
  arr[i] = x+i;
}
```

⟹

```
x=a*b
for(int i=0; i<n; i++)
{ arr[i] = x+i;
}
```

Loop unrolling :-

Repeats the loop body multiple time to reduce loop control instructions and enable more parallelism.

```
for(int i=0; i<4; i++)
{
  a[i] = b[i] + 1;
}
```

⇒

```
a[0] = b[0] + 1 ✓
a[1] = b[1] + 1 ✓
a[2] = b[2] + 1 ✓
a[3] = b[3] + 1 ✓
```

Loop fusion :-

Combines multiple loops that iterate over the same large into a single loop to reduce overhead and improve data locality

```
for(int i=0; i<n; i++)  a[i]= b[i]+1
for(int i=0; i<n; i++)  c[i]= d[i]+1
```




```
for(int i=0; i<n; i++)
{ a[i]= b[i]+1
  c[i]= d[i]+1
}
```

Loop fission :- (Also distribution)

Split a loop into two or more loops to improve parallelism and cache behaviour

```
for(int i=0, i<n; i++)
{ a[i] = b[i] + 1;
  x[i] = y[i] - 1;
}
```

⇒

```
for(int i=0; i<n; i++) a[i] = b[i]+1 ;
for(int i=0; i<n; i++) x[i] = y[i] - 1 ;
```

Strength Reduction :-

Replaces costly operation (e.g multiplication) with cheaper one (i.e addition or left shift

```
for(i=0; i<n; i++)
{ x = i * 4;
}
```

```
for(i=0; i<n; i++)
{
   x = i << 2;
}
```

Loop interchange :-

Swaps nested loops to improve access pattern and cache performance

```
for(int i=0; i<N; i++)
{  for(int j=0; j<m; j++)
   {  A[j][i]=0;
   }
}
```

⇒

```
for(j=0; j<m; j++)
{ for(i=0; i<n; i++)
  {  A[j][i]=0
  }
}
```

THANK - YOU